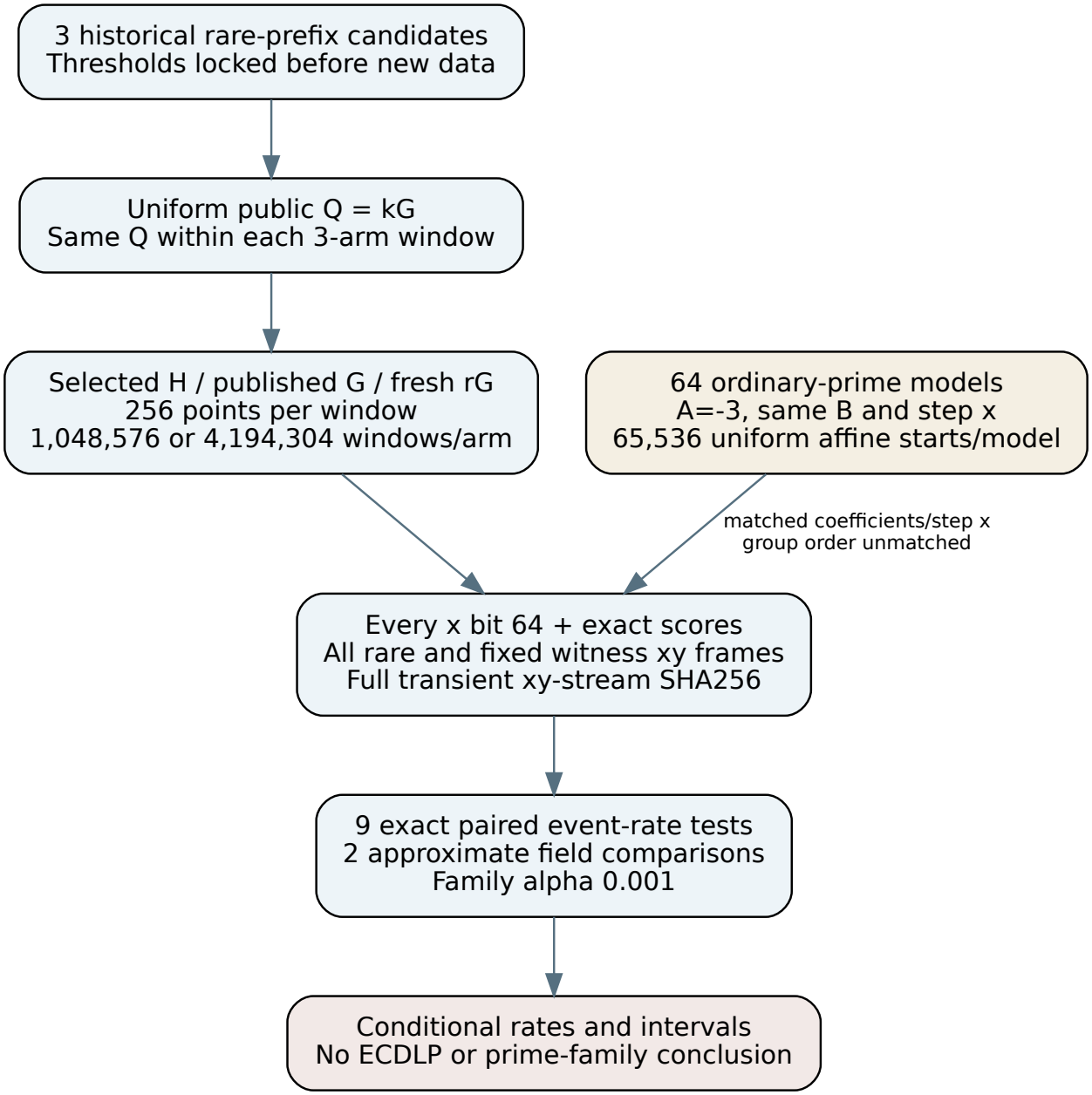
# Powered rare point-pattern study

12 October 2026 | run powered-20261012 | observations only; no scientific-state transition

5,905,580,032 point observations; 23,068,672 256-point window observations; 640/640 jobs; 0 failures. 0/11 predeclared comparisons cross the corrected threshold.

Elapsed wall 5,847.83s; summed measured job CPU 17,497.85s. Host platform: Linux-6.18.44-x86_64-with-glibc2.41. Runtime facts only; no speedup or ECDLP-cost claim.

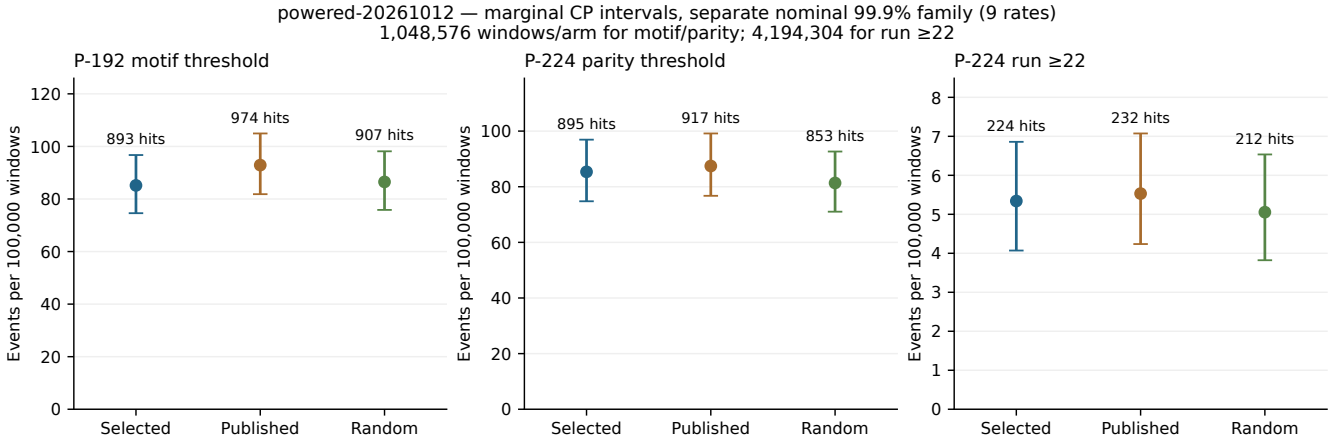
Same uniform Q in selected, actual published, and fresh random-generator cohorts. 64 ordinary-prime models match A=-3, B and step x; group orders are unmatched. Points within each window are dependent.




Locked seed: solinas-rare-events-20261012-powered-v1. Source committed and pushed before launch: 2f58493ab588832a11ac5771e0ba023958c44dfb. Exact parameters and curve UIDs: tasks/solinas_rare_events/inputs.json.

## Event rates and uncertainty

P-192 motif threshold: selected 893/1,048,576 (85.163/100,000); published 974/1,048,576 (92.888/100,000); random 907/1,048,576 (86.498/100,000)
P-224 parity threshold: selected 895/1,048,576 (85.354/100,000); published 917/1,048,576 (87.452/100,000); random 853/1,048,576 (81.348/100,000)
P-224 run ≥22: selected 224/4,194,304 (5.341/100,000); published 232/4,194,304 (5.531/100,000); random 212/4,194,304 (5.054/100,000)

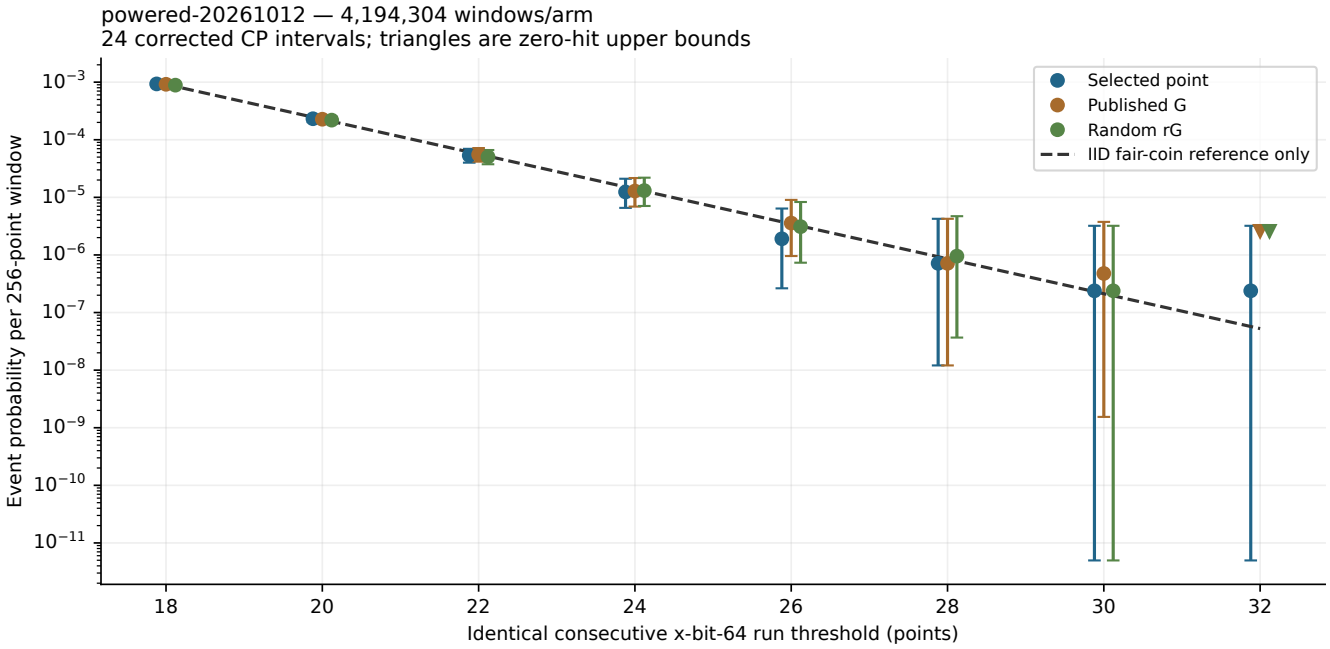



Intervals: separate nine-rate CP family, Bonferroni alpha0.001/9. Primary tests: nine exact paired McNemar comparisons plus two approximate field comparisons; per-test alpha0.001/11. Tests with no crossing do not establish a zero effect; all outputs are scoped to the measured descriptors.

Full counts, discordant pairs, p-values and conservative difference intervals: tasks/solinas_rare_events/runs/powered-20261012/analysis.json. Ordinary comparisons use field means, not pooled point counts.

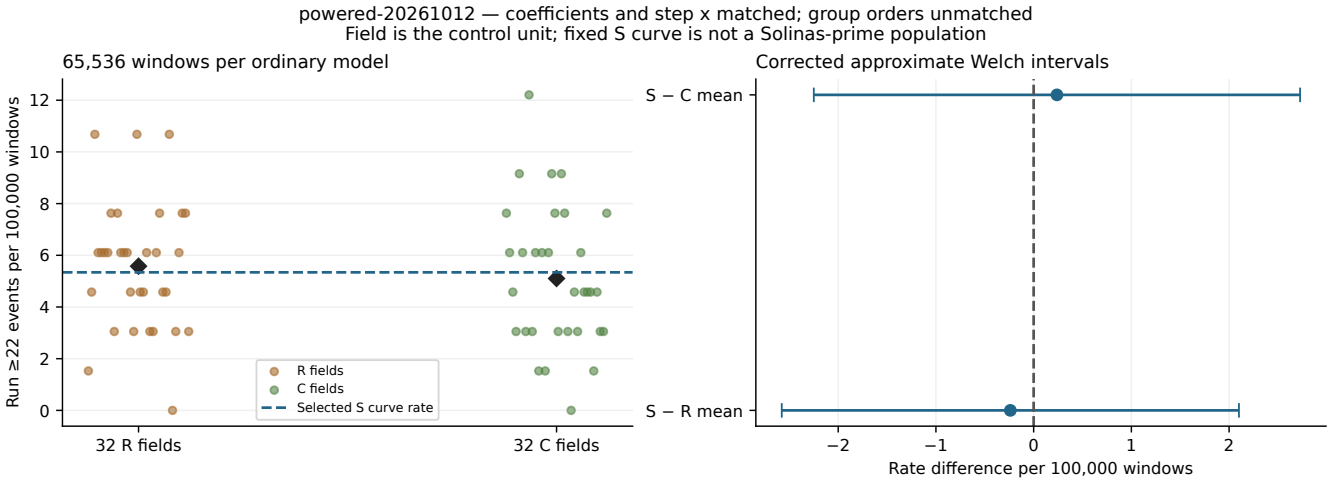
Derived selected/random run≥22 rate ratio: 1.0566; conservative simultaneous-marginal interval [0.6229, 1.7945]. This is conditional on this event and curve/point contrast.

## Long-run tails and field controls



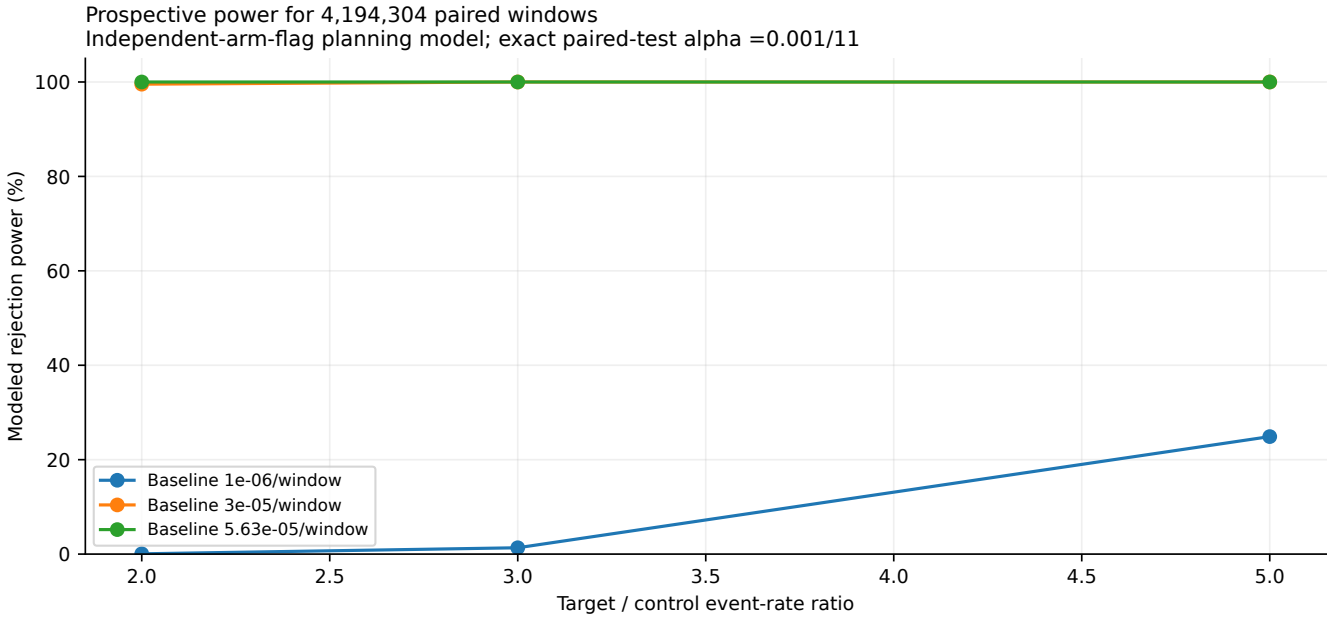

Eight fixed thresholds ×three cohorts; separate24-interval CP family at alpha0.001. Downward triangles mark zero-hit upper bounds. The fair-coin line is a benchmark, not an IID assumption for point bits.

One selected-point window contained32 consecutive zero bits. Maxima: selected32, published31, random30. Threshold32 intervals include the fair-coin reference in all arms; this one hit does not resolve a generator effect. Full raw frames and descriptive start data are preserved.




Each ordinary field has65,536 independent affine starts; 32 fields per pool. The S curve has256 chunks of16,384 starts. Welch intervals are approximate, conditional on a fixed S curve and reused ordinary-prime pools.

## What the experiment can detect




At baseline3e-5 per window, doubling has99.54% modeled power. At baseline1e-6, doubling has only0.057% modeled power. These are prospective independent-arm planning assumptions; actual pair dependence may change power.

Every generator on the prime-order named curve visits the same full point set; it changes the walk order. Consecutive-bit events concern that ordering. Solinas sparsity alone does not establish ECDLP resistance or a solver advantage.

Historical deterministic outliers still exist. Random-start measurements estimate global event prevalence, not revisiting a specific known prefix. A phenomenon confined to a few starts in a2^192 or2^224 group is essentially invisible to a feasible uniform sample. Sample size follows prevalence, effect size, uncertainty and power, not a fixed field-coverage percentage.

The primary, marginal-rate and tail families are separate. Their union is nominal99.7%, subject to Welch approximation. No result excludes all alternate-bit, alternate-coordinate, higher-order or substantially rarer effects. Future work: fresh-seed replication or a separately frozen structured-start study.

## Verification, retention and reproduction

Native online checks: 5,905,580,032 finite on-curve point observations. Separate Python: 23,068,672 score-replayed windows; 6,185,472 coordinate checks; 6,161,310 affine transitions; 8,886 starts; 15,276 scalar multiplications; 6,137,148 x-triple-law checks. No separate research reviewer has signed these observations.

Retained: every measured bit string and exact score; public random-generator scalars; complete rare/witness coordinate frames; per-file and complete transient coordinate-stream hashes. Most non-event full coordinates are regenerated from seeds, rather than stored. Full-coordinate mode is available prospectively for a large filesystem.

Lossless compact archive: solinas-rare-events-20261012-data.zip; 1,338,064,750 bytes; 4,256 entries independently checked after compression. SHA256: 68a67d9912d3fcfee76af151d5d19aa381b216f928ca19d82f3990e783be3019. Earlier datasets and reports are immutable. No cloud upload is asserted.

Reproduce with research.py --output /large-disk/study, then verify.py and analyze.py. No agent, key or service credential is needed. Choose a new public seed and new directory to accumulate a new fixed run. Changed source/config on resume and corrupt artifacts are refused; failed attempts are preserved.

Runtime: Python3.12.14, NumPy2.3.5, SciPy1.17.0, OpenSSL 3.5.7 9 Jun 2026 (Library: OpenSSL 3.5.7 9 Jun 2026); workers3. Exact command/commit/dirty state, source copies, compiler logs and CPU/wall facts are retained; unexposed model/usage metadata is null. Runtime facts are not attack-cost or performance claims.

Evidence: tasks/solinas_rare_events/{PROTOCOL.md,inputs.json,power-plan.json}; runs/powered-20261012/{design.json,completion.json,analysis.json,verification.json}; publication/{dataset.json,dataset-manifest.json,records.jsonl.gz}. Prior evidence: PR2107, source ef9e09d96fa845f0a1614a421e47a5a25b950256. Sources are internal artifacts read in this session; no external literature novelty claim. Canonical ECDLP frontier and prior graph inventory checked; no affected mechanism or official scientific status to change.